Weather Trend Forecasting

PM Accelerator Mission:

By making industry-leading tools and education available to individuals from all backgrounds, **we level the playing field for future PM leaders.** This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, **surround you with the right PM ecosystem**, and discover the new world of AI product management skills.

Project Objective:

Analyze the "Global Weather Repository.csv" dataset to forecast future weather trends and showcase data science skills through both basic and advanced techniques. This dataset provides Daily weather information for cities around the world. This dataset offers a comprehensive set of features that reflect the weather conditions worldwide. It includes over 40 features.

Dataset

- The dataset is available on the Kaggle website.

World Weather Repository: https://www.kaggle.com/datasets/nelgiriyewithana/global-weather-repository/

Handling Missing Values and Outliers

- **Missing Values**: The dataset was checked for missing values, and no significant missing data was found.
- **Outliers**: Outliers were identified and removed using the Interquartile Range (IQR) method. The IQR method was applied to numerical columns such as temperature, wind speed, and precipitation to ensure the data was clean and ready for analysis.

Data Normalization

- The numerical features were normalized using the MinMaxScaler to ensure that all features were on the same scale, which is essential for accurate model training and evaluation

Exploratory Data Analysis (EDA)

Basic EDA

- **Temperature and Precipitation Trends**: Basic visualizations were created to explore trends in temperature and precipitation over time. Line plots and bar plots were used to visualize the average temperature and precipitation by country and over time.
- **Correlation Analysis**: A correlation matrix was generated to understand the relationships between different weather parameters, such as temperature, humidity, and air quality indices.

Advanced EDA

- **Anomaly Detection**: Outliers were identified and analyzed using the IQR method. This helped in understanding extreme weather conditions and their potential impact on the dataset.
- **Geographical Patterns**: Geographical patterns in temperature and humidity were visualized using scatter plots on a world map. This helped in identifying regions with extreme weather conditions.

Model Building

Forecasting Models

Three different forecasting models were built and evaluated to predict future temperature trends:

1. **Exponential Smoothing (ETS)**:
    - The ETS model was used to capture trends and seasonality in the temperature data.
    - **Metrics**:
        - MAE: 0.19
        - MSE: 0.04
        - RMSE: 0.21
2. **Prophet**:
    - The Prophet model, developed by Facebook, was used to forecast temperature trends, taking into account seasonality and holidays.
    - **Metrics**:

        - MAE: 2.18
        - MSE: 6.21
        - RMSE: 2.4

3. **ARIMA**:
    - The ARIMA model was used to capture the autoregressive and moving average components of the temperature data.
    - **Metrics**:
        - MAE: 0.16
        - MSE: 0.03
        - RMSE: 0.19

Ensemble Model

- An ensemble model was created by combining the predictions from the ETS, Prophet, and ARIMA models using a weighted average.
- **Metrics**:
    - MAE: 0.46
    - MSE: 0.25
    - RMSE: 0.50

The ensemble model performed slightly better than the individual models, demonstrating the power of combining multiple models to improve forecast accuracy.

Advanced Analyses

Climate Analysis

- **Long-Term Trends**: Long-term trends in temperature and precipitation were analyzed by aggregating data by year and country. The analysis showed a gradual increase in global temperatures over the years, with some regions experiencing more significant changes than others.
- **Seasonal Patterns**: Seasonal patterns in temperature and precipitation were analyzed by aggregating data by month and country. The analysis revealed clear seasonal trends, with higher temperatures and precipitation during certain months.

Environmental Impact

- **Air Quality Trends**: Air quality trends were analyzed by resampling the data to monthly averages. The analysis showed fluctuations in air quality indices over time, with some pollutants showing seasonal patterns.
- **Correlation with Weather Parameters**: A correlation analysis was performed to understand the relationship between air quality and weather parameters such as temperature, humidity, and wind speed. The analysis revealed that certain pollutants, such as PM2.5 and PM10, were positively correlated with temperature and humidity.

Spatial Analysis

- **Geographical Distribution**: The geographical distribution of temperature and humidity was visualized using scatter plots on a world map. The analysis showed that certain regions, such as the equator, consistently experienced higher temperatures, while others, such as the poles, had lower temperatures.
- **Heatmaps**: Heatmaps were created to visualize the concentration of temperature and humidity across different regions. This helped in identifying hotspots with extreme weather conditions.

Insights and Conclusion

Key Insights

1. **Global Warming**: The analysis revealed a gradual increase in global temperatures over the years, with some regions experiencing more significant changes than others.
2. **Seasonal Patterns**: Clear seasonal patterns were observed in temperature and precipitation, with higher temperatures and precipitation during certain months.
3. **Air Quality**: Air quality indices showed fluctuations over time, with certain pollutants showing seasonal patterns. The analysis also revealed a positive correlation between air quality and weather parameters such as temperature and humidity.
4. **Geographical Patterns**: The geographical distribution of temperature and humidity showed that certain regions consistently experienced extreme weather conditions.

Conclusion

The project successfully analyzed the "Global Weather Repository.csv" dataset to forecast future weather trends and uncover key insights into global climate patterns. The ensemble model, which combined the predictions from the ETS, Prophet, and ARIMA models, provided the most accurate forecasts. The advanced analyses, including climate analysis, environmental impact, and spatial analysis, provided valuable insights into long-term climate trends, air quality, and geographical patterns in weather conditions.